

BIT3208: ADVANCED WEB DESIGN AND DEVELOPMENT
PRACTICAL LOGBOOK

Student Name: JOSEPH KINUTHIA
Admission Number: BBIT202456504
Course/Class: Bachelor of Business Information Technology
Unit Code: BIT3208
Unit Name: Advanced Web Design and Development
Lecturer's Name: Mr. Nyoro
Semester/Academic Year: YEAR 3 SEM 2, 2026

PROJECT DETAILS

Project Title: Smart E-Commerce Web Application

Selected Technologies:

- **Frontend:** HTML5, CSS3, JavaScript
- **Backend:** PHP 8.2
- **Database:** MySQL (MariaDB)
- **Server:** XAMPP (Apache)
- **Version Control:** Git & GitHub

ONLINE PORTFOLIO / GITHUB

GitHub Repository: <https://github.com/joeflinch31/smart-ecommerce-capstone>

Local Development: <http://127.0.0.1/week1/>

TABLE OF CONTENTS

1.0 Week 1: Local Environment Setup	3
1.1 Installation of XAMPP	3
1.2 Testing Localhost	4
1.3 HTML Hello World Page	5
1.4 PHP Hello World Page	6
1.5 Database Connection Test	7
1.6 Week 1 Reflection	8
1.7 Week 1 Checklist	9

WEEK 1: LOCAL ENVIRONMENT SETUP

Title: Installation and Testing of Local Development Environment

Fig 1.1: XAMPP Installation Directory

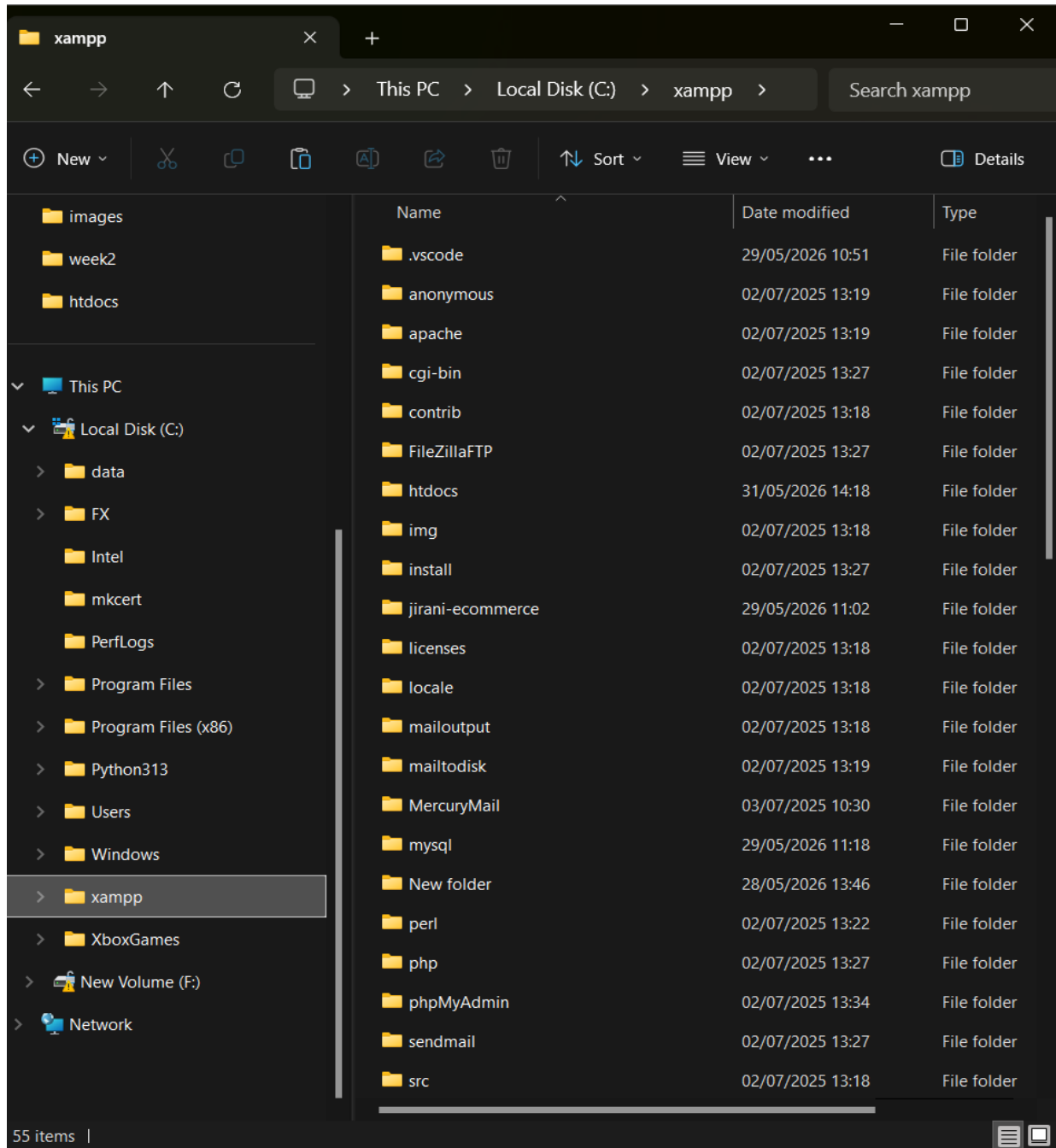


Figure 1.1 shows the XAMPP installation directory at C:\xampp containing all required subfolders including Apache, MySQL, PHP, phpMyAdmin, andhtdocs.

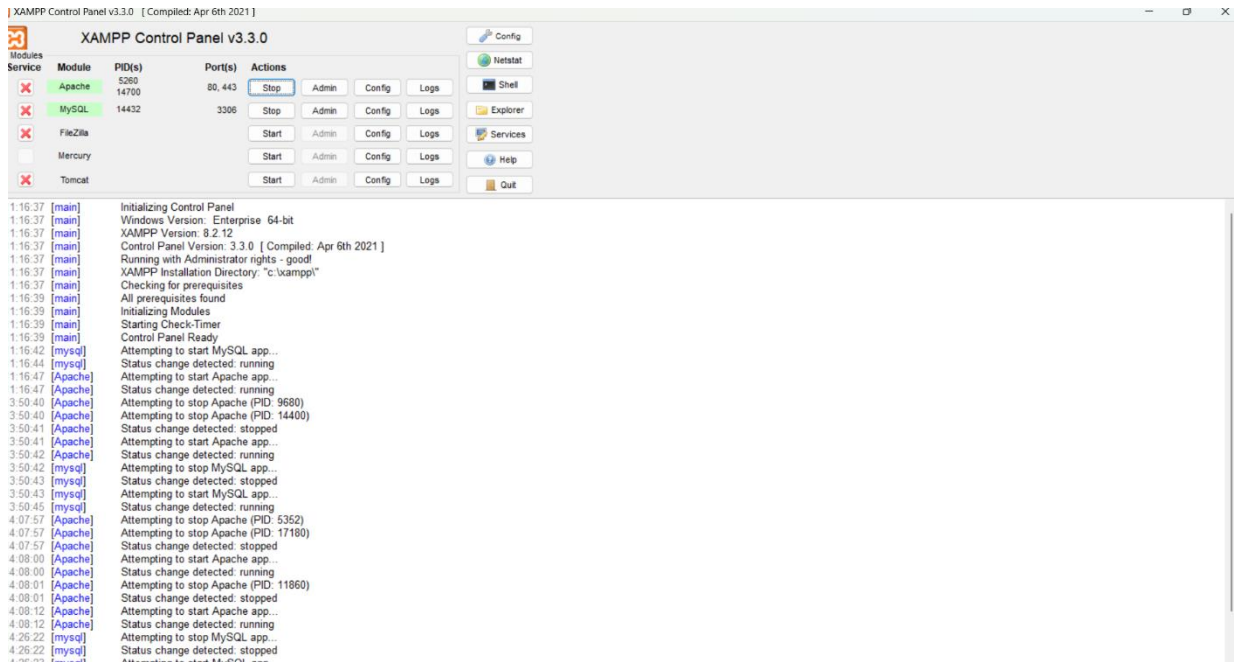


Figure 1.2 shows the XAMPP Control Panel with Apache and MySQL running (green status).

Fig 1.3: Localhost Server Test

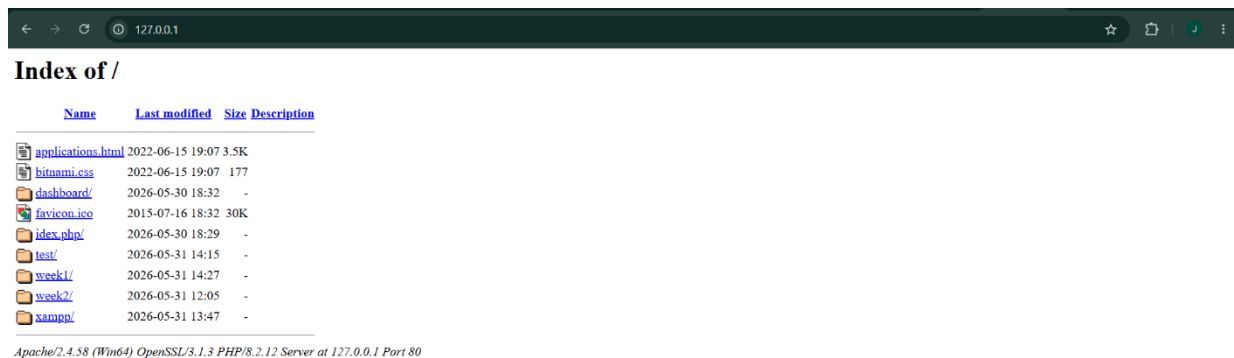


Figure 1.3 shows the browser displaying http://127.0.0.1/ with the week1 folder visible.

Fig 1.4: Week1 Project Folder Structure

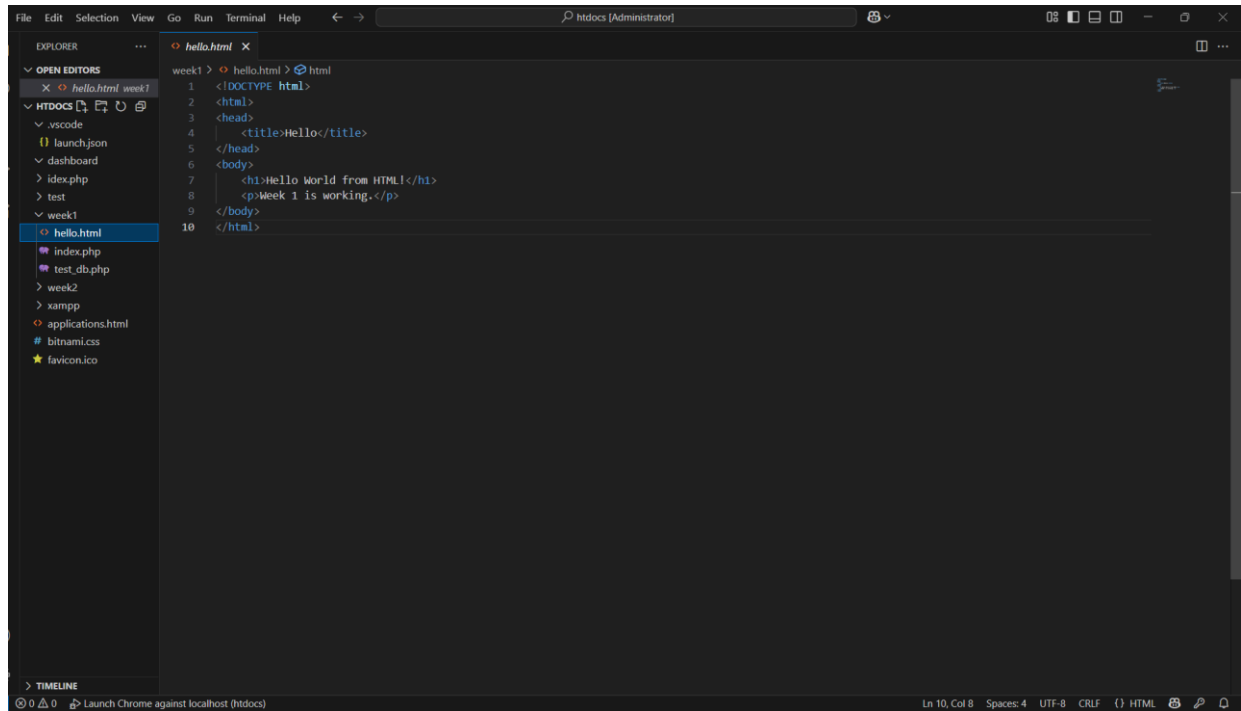
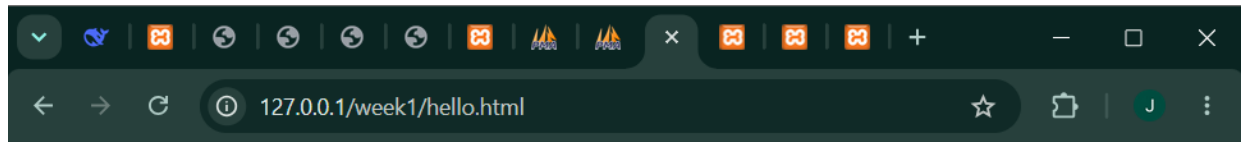


Figure 1.4 shows the Week1 folder containing hello.html, index.php, and test_db.php.

Fig 1.5: HTML Hello World Page



Hello World from HTML!

Week 1 is working.

Figure 1.5 shows the HTML Hello World page at `http://127.0.0.1/week1/hello.html` displaying "Hello World!".

Fig 1.6: PHP Hello World Page

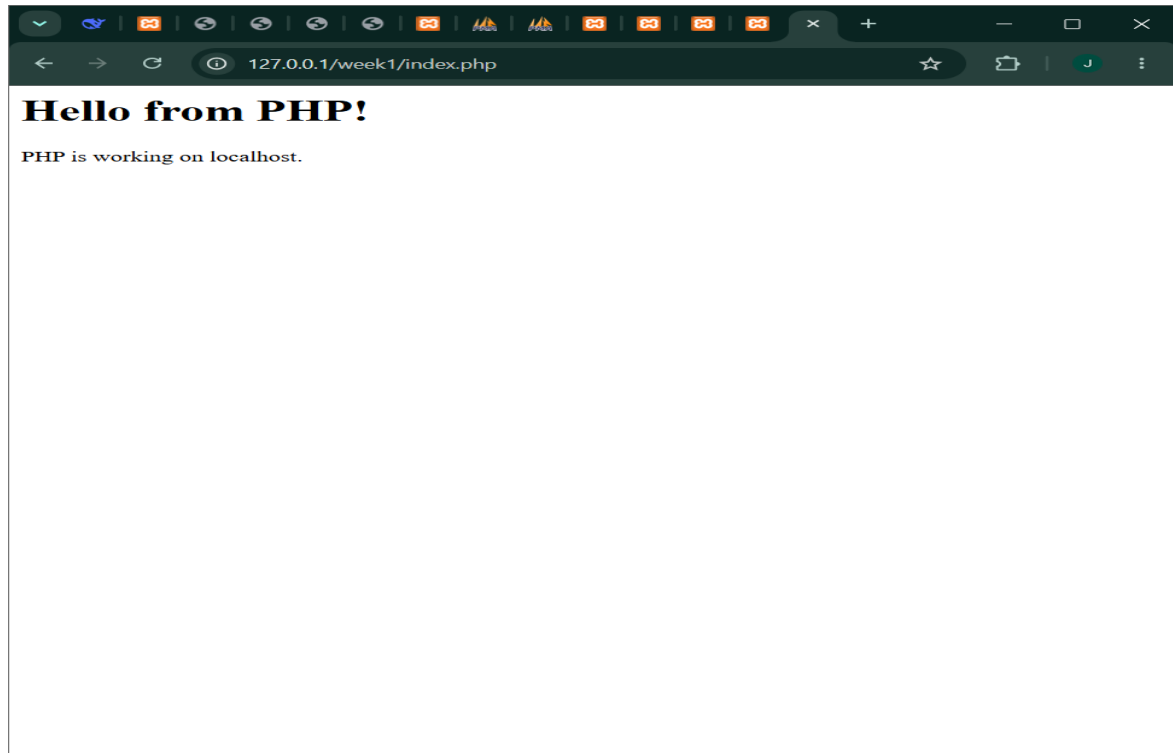


Figure 1.6 shows the PHP Hello World page at `http://127.0.0.1/week1/index.php` displaying "Hello from PHP!".

Fig 1.7: Database Connection Test

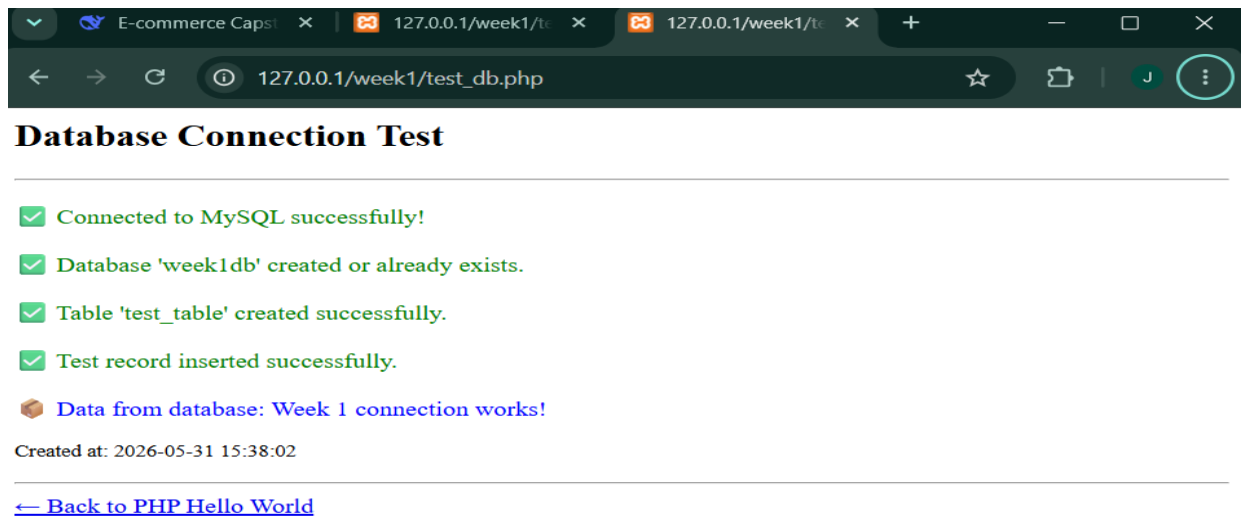


Figure 1.7 shows the database test with green checkmarks confirming MySQL connection, database creation, table creation, and data insertion.

Fig 1.8: phpMyAdmin Database Verification

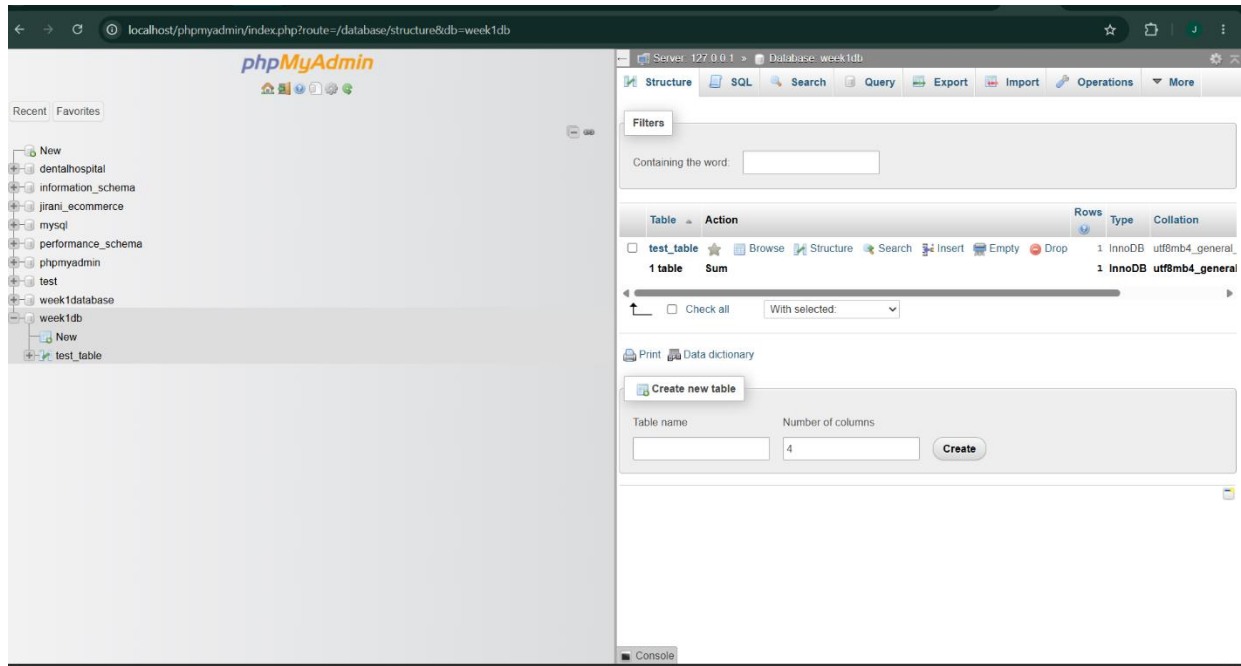


Figure 1.8 shows phpMyAdmin with week1db database and test_table containing the test record.

WEEK 1 REFLECTION

During Week 1, I successfully set up my local development environment using XAMPP 8.2.12 with Apache and MySQL services. I created three test files: hello.html (pure HTML), index.php (PHP processing), and test_db.php (database connection script). The PHP script connects to MySQL, creates database week1db, creates test_table, inserts a test record, and reads it back - all confirmed through phpMyAdmin. The main challenge was understanding the document root location (C:\xampp\htdocs) where all project files must be placed. I learned the difference between static HTML (client-side) and PHP (server-side) execution.

WEEK 1 COMPLETION CHECKLIST

Task	Status
------	--------

XAMPP installed	✓ Complete
Apache server running	✓ Complete
MySQL server running	✓ Complete
Localhost accessible	✓ Complete
hello.html created	✓ Complete
index.php created	✓ Complete
test_db.php created	✓ Complete
Database week1db created	✓ Complete
Table test_table created	✓ Complete
Test record inserted	✓ Complete
Database exported	✓ Complete
All screenshots taken	✓ Complete